



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No. 5

Title : Apply Array Methods and Object Handling; Create a Cart Total Calculator with Discount Logic

Aim : To implement a shopping cart total calculator using JavaScript arrays and objects, apply array methods for processing cart data, and calculate the final payable amount using discount logic.

Software Tools Required

- Visual Studio Code
- HTML5
- CSS3
- JavaScript (ES6)
- Google Chrome or any modern web browser

Theory

The shopping cart is implemented using an array of objects, where each object stores the item name, price, and quantity. When the user enters product details and clicks the Add Item button, a new object is created and added to the cart using the **push() method**.

The **displayCart()** function uses the **forEach()** method to traverse the array and display all cart items in a table. Each item's total is calculated as price × quantity, and a Remove button is provided for deleting individual items.

The **calculateCart()** function uses **map()** to create an array of item totals and **reduce()** to calculate the subtotal. Conditional statements apply a 20% discount for subtotals of ₹5000 or more and a 10% discount for subtotals of ₹3000 or more. The final amount is obtained after subtracting the discount and is displayed dynamically using DOM manipulation.

The **splice()** method removes selected items from the cart, while the Clear Cart option empties the array. After every operation, the table and summary are refreshed to keep the displayed information up to date.

Conclusion

The experiment demonstrates the use of JavaScript object handling and array methods such as **push()**, **forEach()**, **map()**, **reduce()**, and **splice()** to build an interactive cart total calculator with dynamic updates and discount calculation.



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